12TH BIOLOGY · CHAPTER 20 · SOLVED EXERCISE

Biotechnology

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. Identify the FALSE statement.

(a) Restriction enzyme is used for the production of RFLPs.
(b) DNA ligase enzyme is used to cut DNA, creating the sticky ends.
(c) Reverse transcriptase enzyme is used for making cDNA from mRNA.
(d) Electrophoresis means the separation of DNA fragments.

Answer: (b) DNA ligase enzyme is used to cut DNA, creating the sticky ends. — DNA ligase joins DNA fragments (molecular glue). DNA is cut, with sticky ends, by restriction endonucleases.

2. Which bacterium is used in the production of insulin by genetic engineering?

(a) Saccharomyces
(b) Rhizobium
(c) Escherichia
(d) Mycobacterium

Answer: (c) Escherichia — The recombinant plasmids are introduced into cultures of *Escherichia coli*.

3. Which technique is used for cloning of gene/DNA?

(a) RFLP analysis
(b) Polymerase Chain Reaction (PCR)
(c) Electroporation
(d) Gel electrophoresis

Answer: (b) Polymerase Chain Reaction (PCR) — PCR makes multiple identical copies of the gene of interest.

4. Which method is used for obtaining the gene of interest?

(a) Artificial gene synthesis
(b) Reverse transcription
(c) Cleavage from chromosome
(d) All of these

Answer: (d) All of these — The textbook lists all three ways of obtaining the gene of interest.

5. When a gene of interest is attached to the vector, the new molecule is called;

(a) Recombinant DNA
(b) Restriction length fragment
(c) Complimentary DNA
(d) Genomic library

Answer: (a) Recombinant DNA — The vector carrying the gene of interest becomes recombinant DNA (rDNA).

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6. What is the use of plasmid in genetic engineering?

(a) To carry DNA to a cell (b) To cut DNA
(c) To join pieces of DNA (d) To replicate DNA

Answer: (a) To carry DNA to a cell — A plasmid is a vector (molecular carrier) that introduces the gene of interest into the host cell.

7. Which of these can serve as ideal expression system for gene cloning?

(a) Bacteriophages (b) Bacterial cells
(c) Plant cells (d) Animal cells

Answer: (b) Bacterial cells — Bacterial cells have a short generation time and a simple genetic system.

8. In PCR, optimum temperature for carrying out the extension of primer is;

(a) 98° C. (b) 72° C.
(c) 60° C (d) 37° C.

Answer: (b) 72° C. — 98°C is for denaturation and 60°C for annealing; extension is best at 72°C.

9. Which enzyme is used for the extension of primer in PCR?

(a) DNA polymerase (b) Transcriptase
(c) DNA ligase (d) Restriction endonuclease

Answer: (a) DNA polymerase — The heat-stable DNA polymerase called Taq polymerase adds nucleotides to the 3′ end of the primer.

10. What is true for a Genetically Modified Organism

(a) Genes have been introduced (b) Genes have been enhanced
(c) Genes have been deleted (d) All of these

Answer: (d) All of these — A GMO is an organism whose genome has been changed; genes may be inserted, deleted or modified.

MCQ 1 uses the terms RFLP, cDNA and electrophoresis, which are not explained in the body of this chapter. In MCQ 2, the label of option (b) is printed as "b)".

SECTION 2: SHORT QUESTIONS

25. What is the role of restriction endonucleases and DNA ligases in gene cloning?

Restriction endonucleases (molecular scissors) cut DNA at specific sequences called restriction sites. They cut out the gene of interest and open the vector, leaving compatible sticky ends on both. **DNA ligase** (molecular glue) then joins the gene of interest with the cleaved vector by forming phosphodiester linkages, which makes the recombinant DNA.

26. What is the role of vectors in recombinant DNA technology?

Vectors are molecular carriers. The gene of interest is attached to the vector, which becomes recombinant DNA, and this rDNA is introduced into the host cells. So the vector carries the gene of interest into the host. Bacterial plasmids are the most commonly used vectors; others are bacteriophages, cosmids, BACs and YACs.

27. Describe how the gene of interest is selected and isolated during recombinant DNA technology.

The gene of interest is the gene which is to be manipulated. It is obtained by direct synthesis in the lab (with polymerases and ligases), by reverse transcription of its mRNA (with reverse transcriptase), or by cleavage from the chromosome. For isolation, the source DNA is cleaved with a restriction endonuclease, which cuts out the gene with sticky ends.

28. State the steps for the integration of gene of interest into the vector.

1. **DNA cleavage:** The source DNA is cut with a restriction endonuclease to isolate the gene of interest. The vector (e.g., plasmid) is cut with the same endonuclease, so both have compatible sticky ends.
2. **Production of recombinant DNA:** The cleaved vector and the gene of interest are incubated together with DNA ligase. It joins them by phosphodiester linkages and the vector becomes a recombinant vector.

29. State the role of GMO in making biotechnology products.

GMOs act as living factories that make the product of the inserted gene.

- **Transgenic bacteria:** medicines (insulin, clotting factors, vaccines, growth hormone, interferon), food enzymes (alpha-amylase, chymosin) and enzymes that degrade pollutants.
- **Transgenic animals:** useful proteins in milk, e.g., human lysozyme and an anticoagulant drug from goats.
- **Transgenic plants:** insect-resistant and herbicide-resistant crops, golden rice, and biopharmaceuticals (antibodies, vaccines).

30. Differentiate between;

• Endonuclease and ligase

Endonuclease	Ligase
It cuts DNA at specific nucleotide sequences (restriction sites).	It joins two DNA fragments.
It is called molecular scissors.	It is called molecular glue.
It breaks the DNA strands and produces sticky ends.	It forms phosphodiester linkages between adjacent nucleotides.

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• PCR and Recombinant DNA Technology

PCR	Recombinant DNA Technology
It rapidly amplifies a tiny segment of DNA into a large quantity.	It cuts DNA from one organism and “pastes” it into the DNA of another.
It is done in vitro; no host cell is needed.	The recombinant DNA is introduced into a host (e.g., bacteria).
It needs primers, nucleotides and Taq polymerase.	It needs restriction endonucleases, a vector, DNA ligase and an expression system.
Product: multiple identical copies of a DNA fragment.	Product: new genetic combinations or specific proteins.

SECTION 3: LONG QUESTIONS

32. Describe the components and steps of recombinant DNA technology.

Recombinant DNA technology means cutting, splicing and combining DNA molecules from different sources into a single hybrid molecule. It also involves introducing the recombinant DNA (rDNA) into a host (e.g., bacteria) to produce new genetic combinations or specific proteins.

Components (tools)

1. Gene of interest

The gene of interest is the gene which is to be manipulated. It can be obtained in three ways:

1. Direct synthesis of the gene in the lab by using enzymes such as polymerases and ligases.
2. Reverse transcription of its mRNA by using reverse transcriptase enzymes.
3. Cleavage from the chromosome by using restriction endonucleases.

2. Restriction endonucleases

These enzymes (molecular scissors) cut DNA at specific nucleotide sequences called restriction sites and leave sticky ends.

3. Vectors

Vectors (molecular carriers) are special molecules used to introduce the gene of interest into host cells. The gene of interest is attached to the vector, which becomes a recombinant DNA (rDNA); the rDNA is then introduced into the host cells. Bacterial **plasmids** are the most commonly used vectors. Other vectors are:

- Bacteriophages (e.g., phage lambda, bacteriophage P1)
- Cosmids (plasmids that contain a small region of bacteriophage DNA called the cos sequence)
- Bacterial artificial chromosomes (BACs)
- Yeast artificial chromosomes (YACs)

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- P1 artificial chromosomes (features of both P1 vectors and BACs)
- Animal viruses (e.g., human cold virus), which carry genes into monkey and human cells

4. DNA ligase

DNA ligase joins two DNA fragments, so it is called **molecular glue**. It forms the phosphodiester linkage between two adjacent nucleotides.

5. Expression system

An **expression system** is a suitable organism that can act as host for the rDNA and can express the gene of interest. Its selection depends on the type of vector used. Bacterial cells are an ideal expression system because of their short generation time and simple genetic system.

Steps

1. Formation of recombinant vector (plasmid)

DNA cleavage: The source DNA is cleaved with a restriction endonuclease and the gene of interest is isolated. The DNA of the vector (e.g., plasmid) is cleaved with the same endonuclease. So compatible sticky ends are produced on both.

Production of recombinant DNA: The cleaved vector and the gene of interest are incubated together in the presence of DNA ligase. It joins them by phosphodiester linkages, and the vector becomes a recombinant vector.

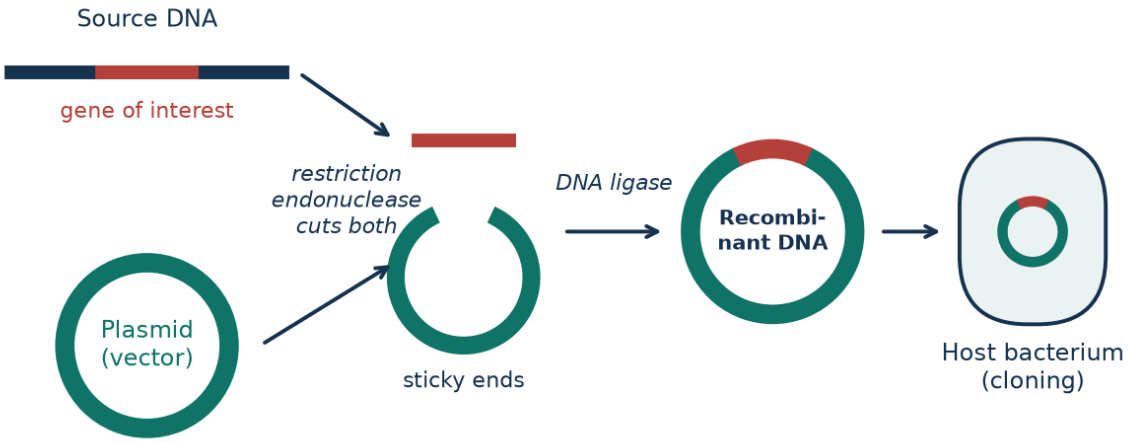
2. Cloning of recombinant vector

The recombinant DNA and the expression system (usually bacteria) are incubated in the same medium. The bacterial cells take up the recombinant DNA. The bacteria multiply into clones, which also contain identical copies (clones) of the gene of interest.

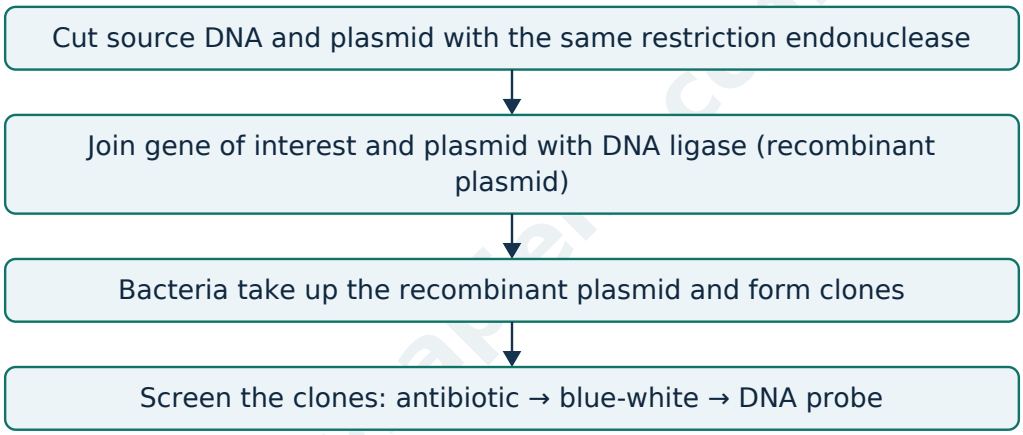
3. Screening of clone library

1. **Selecting for the vector (antibiotic resistance):** The plasmid contains an antibiotic resistance gene. The bacteria are grown on a medium containing that antibiotic (e.g., ampicillin). Only the cells that took up a plasmid survive and grow.
2. **Selecting for the gene of interest (blue-white screening):** The lacZ′ gene of the plasmid makes an enzyme which metabolizes a specific sugar and turns it blue. In a non-recombinant plasmid, lacZ′ stays functional and the colony is blue. An inserted gene of interest “breaks” lacZ′, so the desired recombinant clones remain **white**.
3. **Identifying the specific gene (DNA hybridization):** A radioactive probe (a DNA strand complementary to the gene of interest) is used. It hybridizes only to colonies containing the gene. Auto-radiography (photographic film) locates these colonies on the master plate.

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Formation and cloning of recombinant DNA



Steps of recombinant DNA technology

33. Write a detailed note on restriction enzymes.

Restriction endonucleases are enzymes that cut specific sequences of nucleotides in DNA. They are also called **molecular scissors**.

- There are many kinds of restriction endonucleases. Each kind recognizes a specific nucleotide sequence in DNA, called the **restriction site** or recognition region.
- The enzyme binds to the DNA at that sequence and cuts it at a particular place within the site.
- Restriction sites have **palindromic sequences** of four to eight nucleotides: the two strands have the same nucleotide sequence running in opposite directions.
- The enzyme cleaves both strands. After cleavage, both DNA fragments have single-stranded ends, a few nucleotides long, called **sticky ends**; they are complementary to each other.
- **Value:** any two DNA fragments produced by the same restriction endonuclease can be joined together. So the source DNA and the vector are cut with the same enzyme.

Uses in recombinant DNA technology: they cleave the gene of interest from the chromosome, and they cleave the vector (e.g., plasmid). Because the same enzyme is

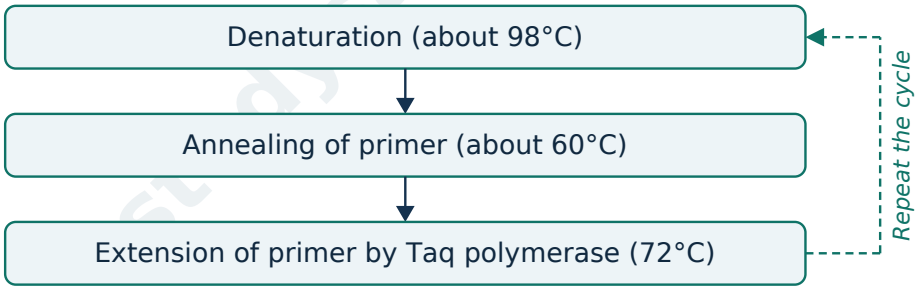
used for both, compatible sticky ends are produced, and DNA ligase can join them into recombinant DNA.

34. Write a note on polymerase chain reaction.

In the **polymerase chain reaction (PCR)**, multiple identical copies of the gene of interest are made. The enzyme DNA polymerase polymerizes a given piece of DNA again and again. Kary Mullis developed PCR in 1983 and won the Nobel Prize in Chemistry in 1993.

Steps of PCR

1. **Denaturation:** An excessive amount of primer (a sequence of 20 to 30 nucleotides) is mixed with the DNA fragment. The mixture is heated to about 98°C. The double-stranded DNA breaks into single strands, each of which can act as a template.
2. **Annealing of primer:** The solution is cooled to about 60°C. The forward and backward primers anneal (hybridize) to the single-stranded DNA at their complementary regions.
3. **Extension of primer:** A very heat-stable DNA polymerase, **Taq polymerase**, and all four types of nucleotides are added. The enzyme adds new nucleotides to the 3′ end of the primer and copies the rest of the fragment (optimum temperature 72°C). Now there are two copies of the original DNA fragment.
4. **Repeating the cycle:** Steps 1 to 3 are repeated and the two copies become four. No more polymerase is needed, because heating does not harm this enzyme. Each cycle takes one or two minutes and doubles the number of DNA molecules. After 20 cycles, a single fragment gives more than one million copies.



One cycle of PCR

Applications of PCR

1. It is an efficient diagnostic technique for detecting the genotypes of infectious agents.
2. It detects genetic mutations responsible for certain genetic diseases and cancers.
3. In criminal investigations, “DNA fingerprints” are prepared from a tiny sample of dried blood or from the base of a single hair.
4. Physicians detect genetic defects in very early embryos by amplifying the DNA of a few sloughed-off cells.
5. It is used to examine the DNA of extinct species, if even a minuscule amount of their DNA remains intact.

6. It is also used to build a genomic library (the collection of the total DNA of a single organism).

35. Describe the methods for the introduction of human DNA into bacteria.

Human DNA (the gene of interest) is introduced into bacteria by recombinant DNA technology, with a plasmid as the vector.

1. Making the recombinant plasmid

The human gene is obtained by direct synthesis, by reverse transcription of its mRNA, or by cleavage from the chromosome. The gene and the plasmid are cut with the same restriction endonuclease and joined by DNA ligase.

2. Putting the recombinant plasmid into bacteria

The recombinant plasmids and the bacteria are incubated in the same medium, and the bacterial cells take up the recombinant DNA. Other vectors, such as bacteriophages and cosmids, can also carry the gene into bacteria. In general, DNA is transferred into a target cell through its wall or membrane without destroying the cell, by these techniques:

Method	How the gene enters the target cell
Electroporation	Electric current creates pores in the cell wall or membrane; the gene enters; the pores close when the current is discharged.
Microinjection	A fine needle injects the gene directly into the nucleus.
Biolistic transformation	The gene, coated with heavy metal ions, is shot into the cell by mechanical force, e.g., gene gun.
Protoplast mediated technique	The cell wall and cell membrane are removed; the protoplast is mixed with a medium containing the gene.
Chemical methods	Naked DNA is fused with a chemical agent; the complex enters the cytoplasm by diffusion or endocytosis.
Virus mediated gene transfer	Adenovirus, adeno-associated virus, HIV, lentivirus and other retroviruses carry the gene.

3. Selecting the bacteria that received the human gene

1. **Selecting for the vector (antibiotic resistance):** The plasmid contains an antibiotic resistance gene. The bacteria are grown on a medium containing that antibiotic (e.g., ampicillin). Only the cells that took up a plasmid survive and grow.
2. **Selecting for the gene of interest (blue-white screening):** The lacZ′ gene of the plasmid makes an enzyme which metabolizes a specific sugar and turns it blue. In a non-recombinant plasmid, lacZ′ stays functional and the colony is blue. An inserted gene of interest “breaks” lacZ′, so the desired recombinant clones remain **white**.
3. **Identifying the specific gene (DNA hybridization):** A radioactive probe (a DNA strand complementary to the gene of interest) is used. It hybridizes only to colonies containing the gene. Auto-radiography (photographic film) locates these colonies on

the master plate.

Example: human insulin gene in bacteria

The production of human insulin was the first major triumph of genetic engineers. The two polypeptide chains are produced separately in bacteria and then joined chemically.

1. **Preparation of the two genes:** Two different DNA sequences are synthesised, one for Chain A and one for Chain B.
2. **Construction of recombinant plasmids:** The DNA for Chain A is inserted into one set of plasmids and the DNA for Chain B into another. The insulin gene is often attached next to a bacterial gene (like lacZ), which helps the bacteria to recognize and express the foreign DNA.
3. **Transformation of host cells:** The recombinant plasmids are introduced into two separate cultures of *Escherichia coli*. Culture 1 is the factory for Chain A and culture 2 for Chain B.
4. **Fermentation and protein expression:** The bacteria are grown in large industrial bioreactors. As they multiply, they read the human DNA and produce large quantities of the insulin chains.
5. **Extraction and chemical joining:** The bacteria are harvested and the chains are extracted and purified. Chain A and Chain B are mixed and linked by di-sulfide bonds between specific amino acids. This gives functional human insulin, identical to that of a healthy pancreas.

INQUISITIVE QUESTIONS

13. Find and describe the accomplishments of the renowned genetic engineers working in private and public institutions in your province or city.

This is an activity. Visit or search the websites of the biotechnology institutes of your province and write about their scientists. A sample answer for Punjab:

- **National Institute for Biotechnology and Genetic Engineering (NIBGE), Faisalabad:** its scientists work on cotton leaf curl virus and on improved crop varieties. Dr. Kauser Abdulla Malik was its founding director; Dr. Shahid Mansoor is known for work on cotton leaf curl disease.
- **Centre of Excellence in Molecular Biology (CEMB), University of the Punjab, Lahore:** founded by Dr. Sheikh Riazuddin; known for work on insect-resistant (Bt) cotton and on genes of inherited diseases. Dr. Tayyab Husnain led work on transgenic cotton there.
- **Other centres:** School of Biological Sciences (Punjab University), Centre of Agricultural Biochemistry and Biotechnology (University of Agriculture, Faisalabad), and the National Institute for Genomics and Advanced Biotechnology (Islamabad).

This information is not in the textbook; it is general knowledge. Confirm names and current posts from the institutes' own websites before writing them in an assignment.

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14. Analyse and interpret the DNA of a child by comparing it with that of two individuals in a case of disputed parenthood.

A child receives half of its DNA from the mother and half from the father. So every band in the child's DNA fingerprint must be present in either the mother or the real father.

1. Take DNA samples from the child, the mother and the two men; amplify the DNA by PCR and prepare the DNA fingerprint (band pattern) of each.
2. Mark the child's bands that match the mother's bands.
3. The remaining bands of the child must have come from the father. Compare them with the two men.

Band of child	Mother	Man 1	Man 2
1	Present	–	–
2	–	Present	–
3	Present	–	Present
4	–	Present	–

Interpretation (sample data): Bands 2 and 4 of the child are not from the mother, and both are present only in Man 1. So Man 1 is the biological father and Man 2 is excluded.

The textbook only mentions that PCR is used to prepare DNA fingerprints; the method and the sample data above are given for explanation.

15. What may be the application of polymerase chain reaction?

1. It is an efficient diagnostic technique for detecting the genotypes of infectious agents.
2. It detects genetic mutations responsible for certain genetic diseases and cancers.
3. In criminal investigations, "DNA fingerprints" are prepared from a tiny sample of dried blood or from the base of a single hair.
4. Physicians detect genetic defects in very early embryos by amplifying the DNA of a few sloughed-off cells.
5. It is used to examine the DNA of extinct species, if even a minuscule amount of their DNA remains intact.
6. It is also used to build a genomic library (the collection of the total DNA of a single organism).

16. The human genome project is regarded as the most ambitious project ever undertaken by man. Justify.

- **Huge aim:** to find the complete sequence of about 3 billion nucleotide pairs of human DNA and to locate all human genes (about 20,000).
- **Huge effort:** it ran from 1990 to 2003, cost about 3 billion dollars, and involved scientists of many countries working together.
- **Huge benefits:** it helps to find the genes of inherited diseases, to diagnose them early, to design better drugs and gene therapy, and to study human evolution.

The Human Genome Project is not described in the textbook; this answer is based on general knowledge.

17. Investigate careers that require an understanding of biotechnology and genetic engineering.

- **Medical and pharmaceutical:** making insulin, vaccines and other drugs; genetic counsellor; molecular diagnostic (PCR) laboratory scientist.
- **Agriculture:** plant breeder and crop biotechnologist (insect-resistant and nutrient-rich crops).
- **Forensic science:** DNA fingerprinting in criminal and parentage cases.
- **Food and industry:** fermentation technologist, enzyme and food-product development.
- **Environment:** using microbes to clean up pollutants.
- **Research and teaching:** universities and research institutes; bioinformatics.

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